

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.005, beta.var.scale = 3\n')
```

```
## sdev = 0.005, beta.var.scale = 3
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.6328  0.8394  0.8467  0.8428  0.8533  0.8750

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5600  0.6933  0.6380  0.7800  0.9450
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5450  0.6811  0.6249  0.7644  0.9161
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.01778 0.04556 0.07167 0.23727 0.26667 1.00000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02333 0.04597 0.06944 0.21031 0.21306 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01611 0.05333 0.13611 0.34049 0.64000 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.5389  0.7167  0.7567  0.7443  0.7800  0.8211

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.2994  0.4786  0.4414  0.6061  0.9139
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.2978  0.4711  0.4291  0.5889  0.8250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008333 0.233889 0.447464 0.988333 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.004444 0.022778 0.288129 0.649861 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.3956  0.9928  0.7249  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.4833  0.6128  0.6606  0.6555  0.7017  0.7983

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1639  0.3344  0.3228  0.4717  0.8861
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1667  0.3300  0.3114  0.4550  0.7839
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.09111 0.33425 0.71278 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9983 1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.002778 0.204178 0.389306 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.3261  0.6683  0.5680  0.8217  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.4994 0.6089  0.6644  0.6581  0.7111  0.7961

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09167 0.25389 0.26448 0.40333 0.87000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.24167	0.23777	0.36722	0.78056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1780	0.3283	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6939	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.08399	0.00000	1.00000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3194	0.3468	0.6117	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5161	0.6422	0.6933	0.6877	0.7306	0.8339

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05278	0.17667	0.19932	0.30556	0.87444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.17444	0.18544	0.29153	0.66833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.02279	0.00000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9246	1.0000	0.9432	1.0000	1.0000

Modality 4 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.01076	0.00000	0.54833

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.04984	0.00000	1.00000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5189	0.6933	0.7361	0.7273	0.7756	0.8500

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04056	0.14722	0.18408	0.27500	0.86556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04222	0.14278	0.15651	0.24611	0.61444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.002231	0.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4950	0.7144	0.6577	0.8728	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001678	0.000000	0.784444

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.01206	0.00000	1.00000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5839	0.8739	0.9039	0.8961	0.9278	0.9667

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
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0.00000 0.01944 0.09722 0.13716 0.20889 0.85111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.09889	0.12174	0.19722	0.53889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.01409	0.00000	1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000264	0.0000000	0.4044444

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001747	0.000000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.02036	0.00000	1.00000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6128	0.8033	0.8333	0.8270	0.8594	0.9139

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01167	0.07278	0.11583	0.17375	0.84278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01333 0.07556 0.10190 0.16778 0.58222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0005866 0.0000000 1.0000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2639 0.5233 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0 0 0 0 0 0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.00543 0.00000 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.6717 0.9189 0.9378 0.9297 0.9517 0.9756

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.06722 0.11387 0.16722 0.84333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1 1 1 1 1 1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1 1 1 1 1 1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1 1 1 1 1 1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
1 1 1 1 1 1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01333 0.06778 0.09343 0.15111 0.50667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0004638 0.0000000 1.0000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.03422 0.00000 1.00000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 4.039e-05 0.000e+00 6.333e-02
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.

```

```
0.0000000 0.0000000 0.0000000 0.0008445 0.0000000 1.0000000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6328 0.8394 0.8467 0.8428 0.8533 0.8750
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5600 0.6933 0.6380 0.7800 0.9450
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5450 0.6811 0.6249 0.7644 0.9161
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01778 0.04556 0.07167 0.23727 0.26667 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02333 0.04597 0.06944 0.21031 0.21306 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01611 0.05333 0.13611 0.34049 0.64000 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5389 0.7167 0.7567 0.7443 0.7800 0.8211
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.2994 0.4786 0.4414 0.6061 0.9139
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.2978 0.4711 0.4291 0.5889 0.8250
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.008333 0.233889 0.447464 0.988333 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.004444 0.022778 0.288129 0.649861 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3956 0.9928 0.7249 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.4833 0.6128 0.6606 0.6555 0.7017 0.7983
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1639 0.3344 0.3228 0.4717 0.8861
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1667 0.3300 0.3114 0.4550 0.7839
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.09111 0.33425 0.71278 1.00000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9983 1.0000 1.0000 1.0000 1.0000 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.002778 0.204178 0.389306 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.3261 0.6683 0.5680 0.8217 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4994 0.6089 0.6644 0.6581 0.7111 0.7961
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09167 0.25389 0.26448 0.40333 0.87000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09333 0.24167 0.23777 0.36722 0.78056
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.1780 0.3283 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6939 1.0000 1.0000 0.9998 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.08399 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.3194 0.3468 0.6117 1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5161 0.6422 0.6933 0.6877 0.7306 0.8339
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05278 0.17667 0.19932 0.30556 0.87444
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05556 0.17444 0.18544 0.29153 0.66833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.02279 0.00000 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9246 1.0000 0.9432 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01076 0.00000 0.54833
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.04984 0.00000 1.00000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5189 0.6933 0.7361 0.7273 0.7756 0.8500
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.04056 0.14722 0.18408 0.27500 0.86556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.04222 0.14278 0.15651 0.24611 0.61444
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.002231 0.000000 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4950  0.7144  0.6577  0.8728  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.001678 0.000000 0.784444
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01206 0.00000 1.00000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5839  0.8739  0.9039  0.8961  0.9278  0.9667
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01944 0.09722 0.13716 0.20889 0.85111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02111 0.09889 0.12174 0.19722 0.53889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01409 0.00000 1.00000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000000 0.0000000 0.0000000 0.0000264 0.0000000 0.4044444
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000000 0.0000000 0.0000000 0.001747 0.0000000 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.000000 0.000000 0.02036 0.000000 1.000000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##  0.6128  0.8033  0.8333  0.8270  0.8594  0.9139
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.01167 0.07278 0.11583 0.17375 0.84278
## Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.01333 0.07556 0.10190 0.16778 0.58222
## Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000000 0.0000000 0.0000000 0.0005866 0.0000000 1.0000000
## Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000  0.0000  0.0000  0.2639  0.5233  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      0      0      0      0      0      0
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.000000 0.000000 0.00543 0.000000 1.000000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##  0.6717  0.9189  0.9378  0.9297  0.9517  0.9756
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.01167 0.06722 0.11387 0.16722 0.84333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.01333 0.06778 0.09343 0.15111 0.50667
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000000 0.0000000 0.0000000 0.0004638 0.0000000 1.0000000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.00000 0.00000 0.03422 0.00000 1.00000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000e+00 0.000e+00 0.000e+00 4.039e-05 0.000e+00 6.333e-02
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000000 0.0000000 0.0000000 0.0008445 0.0000000 1.0000000
##
## ‘ ‘ ‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.